

MATA37H3 — Week 04

Integration Techniques II: Trigonometric Integrals, Partial Fractions & the Weierstrass Substitution

Recap and orientation

Last week covered substitution and integration by parts. This week finishes the integration-techniques toolkit with **trigonometric-integral identities** (for products and powers of $\sin$, $\cos$, $\sec$, $\tan$), **partial fractions** (for rational functions), and one powerful “universal” technique, the **Weierstrass substitution**, which converts *any* rational function of $\sin x$ and $\cos x$ into an ordinary rational function of t , solvable by partial fractions.

Why this matters

A $\sec^n x \tan^m x$ integral attacked with the wrong parity choice, or a partial-fraction integral where long division is forgotten because the numerator’s degree is not smaller than the denominator’s, are classic exam traps. The Weierstrass substitution is the technique to reach for when no other pattern applies: it always works, though it is not always the fastest. Units on improper integrals and series (Weeks 5-14) all assume fluency with this full toolkit.

KEY FACT Powers of $\sin x$ and $\cos x$ (OpenStax §3.2)

To integrate $\int \sin^n x \cos^m x \, dx$:

- If **either** exponent is **odd**, peel off one copy of that function, convert the remaining even power using $\sin^2 x + \cos^2 x = 1$, and substitute $u =$ (the other function). For odd m : write $\cos^m x = \cos^{m-1} x \cos x$, convert $\cos^{m-1} x$ (an even power) via $\cos^2 x = 1 - \sin^2 x$, then let $u = \sin x$.
- If **both** exponents are **even**, use the half-angle identities $\sin^2 x = \frac{1 - \cos(2x)}{2}$ and $\cos^2 x = \frac{1 + \cos(2x)}{2}$ to reduce the powers before integrating (repeat if needed).

KEY FACT Powers of $\sec x$ and $\tan x$ (OpenStax §3.2; MAT137 Unit 9)

To integrate $\int \sec^n x \tan^m x \, dx$ use $\frac{d}{dx}[\tan x] = \sec^2 x$, $\frac{d}{dx}[\sec x] = \sec x \tan x$ and $\tan^2 x + 1 = \sec^2 x$:

- If n (the power of secant) is **even**, peel off $\sec^2 x$, convert the rest of the secant power to tangents via $\sec^2 x = 1 + \tan^2 x$, and substitute $u = \tan x$.
- If m (the power of tangent) is **odd**, peel off one $\sec x \tan x$, convert the remaining even tangent power to secants via $\tan^2 x = \sec^2 x - 1$, and substitute $u = \sec x$.
- The integrals $\int \sec x \, dx = \ln |\sec x + \tan x| + C$ and $\int \csc x \, dx = -\ln |\csc x + \cot x| + C$ fit neither pattern and must be memorized (or re-derived with the trick of multiplying

by $\frac{\sec x + \tan x}{\sec x + \tan x}$).

DEFINITION Partial Fraction Decomposition (OpenStax §3.4)

A rational function $\frac{P(x)}{Q(x)}$ can be decomposed into simpler fractions **only when** $\deg P < \deg Q$. If $\deg P \geq \deg Q$, **long divide first**: $\frac{P(x)}{Q(x)} = A(x) + \frac{R(x)}{Q(x)}$ with $\deg R < \deg Q$, and only then decompose $\frac{R(x)}{Q(x)}$. The shape of the decomposition is determined by factoring $Q(x)$:

- A **distinct linear factor** $(x - a)$ contributes $\frac{A}{x - a}$.
- A **repeated linear factor** $(x - a)^k$ contributes $\frac{A_1}{x - a} + \frac{A_2}{(x - a)^2} + \dots + \frac{A_k}{(x - a)^k}$: one term for *every* power from 1 to k .
- An **irreducible quadratic factor** contributes $\frac{Ax + B}{x^2 + px + q}$ (a linear numerator).

Each piece then integrates with $\int \frac{du}{u} = \ln |u| + C$ (linear factors), the power rule (repeated factors), or $\int \frac{du}{u^2 + a^2} = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$ (irreducible quadratics).

KEY FACT The Weierstrass Substitution, a Universal Rational-Trig Technique (Dummit)

For an integrand that is a rational function of $\sin x$ and $\cos x$, the substitution $t = \tan\left(\frac{x}{2}\right)$ converts the entire integrand into an ordinary rational function of t via

$$\sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}, \quad dx = \frac{2dt}{1+t^2}.$$

The resulting integral in t is always solvable by partial fractions. It is a **fallback**: it always works on rational-trig integrands, but is not always the fastest route.

Worked Examples

EXAMPLE 1

Level: Foundational. Odd power of sine

Evaluate $\int \sin^3 x \cos^2 x \, dx$.

Step 1. Identify the powers: $\sin x$ appears to the power 3 (odd) and $\cos x$ to the power 2 (even). The rule for $\int \sin^m x \cos^n x \, dx$ says that when one exponent is odd, we peel off one copy of that odd-power function.

Step 2. Peel off one factor of $\sin x$: $\sin^3 x = \sin^2 x \cdot \sin x$. The reason for keeping a single $\sin x$ is that it will become the du of a substitution.

Step 3. The remaining even power $\sin^2 x$ must be written using cosines. Use the Pythagorean identity $\sin^2 x + \cos^2 x = 1$, which gives $\sin^2 x = 1 - \cos^2 x$.

Step 4. Rewrite the integral:

$$\int \sin^3 x \cos^2 x \, dx = \int (1 - \cos^2 x) \cos^2 x \sin x \, dx.$$

Now every factor except the lone $\sin x$ is a function of $\cos x$.

Step 5. Choose the substitution $u = \cos x$. Differentiating, $du = -\sin x \, dx$, so $\sin x \, dx = -du$.

Step 6. Substitute: $\cos^2 x = u^2$ and $\sin x \, dx = -du$, so the integral becomes

$$-\int (1 - u^2) u^2 \, du.$$

Step 7. Expand the product inside the integral: $(1 - u^2)u^2 = u^2 - u^4$. So we need $-\int (u^2 - u^4) \, du$.

Step 8. Apply the power rule $\int u^k \, du = \frac{u^{k+1}}{k+1}$ to each term:

$$-\left(\frac{u^3}{3} - \frac{u^5}{5}\right) + C.$$

Step 9. Distribute the minus sign: $-\frac{u^3}{3} + \frac{u^5}{5} + C$.

Step 10. Replace u by $\cos x$ to return to the original variable: $-\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C$.

Step 11. Verify by differentiating. By the chain rule, $\frac{d}{dx} \left[-\frac{\cos^3 x}{3} \right] = -\frac{3 \cos^2 x (-\sin x)}{3} = \cos^2 x \sin x$ and $\frac{d}{dx} \left[\frac{\cos^5 x}{5} \right] = \frac{5 \cos^4 x (-\sin x)}{5} = -\cos^4 x \sin x$.

Step 12. Add the two derivatives: $\cos^2 x \sin x - \cos^4 x \sin x = \cos^2 x \sin x (1 - \cos^2 x) = \cos^2 x \sin x \cdot \sin^2 x = \sin^3 x \cos^2 x$, which is the original integrand.

Answer. $-\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C$

EXAMPLE 2

Level: Foundational. Both powers even: half-angle identity

Evaluate $\int \cos^2 x \, dx$.

Step 1. Identify the powers: $\cos x$ appears to the power 2 and $\sin x$ to the power 0. Both are even, so there is no odd factor to peel off and the substitution method of Example 1 is unavailable.

Step 2. Instead, lower the power with a half-angle identity. Recall the double-angle formula $\cos(2x) = 2 \cos^2 x - 1$.

Step 3. Solve that formula for $\cos^2 x$: add 1 to both sides to get $1 + \cos(2x) = 2 \cos^2 x$, then divide by 2:

$$\cos^2 x = \frac{1 + \cos(2x)}{2}.$$

Step 4. Replace the integrand using this identity:

$$\int \cos^2 x \, dx = \int \frac{1 + \cos(2x)}{2} \, dx.$$

Step 5. Pull out the constant factor $\frac{1}{2}$ and split the sum:

$$\frac{1}{2} \int 1 \, dx + \frac{1}{2} \int \cos(2x) \, dx.$$

Step 6. The first integral is $\frac{1}{2} \int 1 \, dx = \frac{x}{2}$.

Step 7. For the second integral use the substitution $w = 2x$, so $dw = 2 \, dx$ and $dx = \frac{dw}{2}$.

$$\text{Then } \int \cos(2x) \, dx = \frac{1}{2} \int \cos w \, dw = \frac{\sin w}{2} = \frac{\sin(2x)}{2}.$$

Step 8. Multiply by the outside factor $\frac{1}{2}$: $\frac{1}{2} \cdot \frac{\sin(2x)}{2} = \frac{\sin(2x)}{4}$.

Step 9. Combine the pieces and add the constant:

$$\frac{x}{2} + \frac{\sin(2x)}{4} + C.$$

Step 10. Verify by differentiating: $\frac{d}{dx} \left[\frac{x}{2} \right] = \frac{1}{2}$ and $\frac{d}{dx} \left[\frac{\sin(2x)}{4} \right] = \frac{2 \cos(2x)}{4} = \frac{\cos(2x)}{2}$.

The sum is $\frac{1 + \cos(2x)}{2} = \cos^2 x$, as required.

Answer. $\frac{x}{2} + \frac{\sin(2x)}{4} + C$

EXAMPLE 3

Level: Intermediate. Both powers even: identity used twice

Evaluate $\int \cos^4 x \, dx$.

Step 1. Identify the powers: $\cos x$ has power 4 and $\sin x$ has power 0, both even, so use the half-angle identity $\cos^2 x = \frac{1 + \cos(2x)}{2}$.

Step 2. Write the fourth power as a square of a square: $\cos^4 x = (\cos^2 x)^2$.

Step 3. Apply the identity inside the parentheses:

$$\cos^4 x = \left(\frac{1 + \cos(2x)}{2} \right)^2.$$

Step 4. Square the fraction: numerator $(1 + \cos 2x)^2 = 1 + 2 \cos 2x + \cos^2 2x$ (using $(a + b)^2 = a^2 + 2ab + b^2$) and denominator $2^2 = 4$. So

$$\cos^4 x = \frac{1 + 2 \cos 2x + \cos^2 2x}{4}.$$

Step 5. The term $\cos^2 2x$ is again an even power, so apply the half-angle identity a second

time. Replace the angle x by $2x$ in $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$ to get $\cos^2 2x = \frac{1 + \cos 4x}{2}$.

Step 6. Substitute this into the numerator:

$$\cos^4 x = \frac{1}{4} \left(1 + 2 \cos 2x + \frac{1 + \cos 4x}{2} \right) = \frac{1}{4} + \frac{\cos 2x}{2} + \frac{1}{8} + \frac{\cos 4x}{8}.$$

Step 7. Combine the constants: $\frac{1}{4} + \frac{1}{8} = \frac{2}{8} + \frac{1}{8} = \frac{3}{8}$. So

$$\cos^4 x = \frac{3}{8} + \frac{\cos 2x}{2} + \frac{\cos 4x}{8}.$$

Step 8. Integrate term by term. $\int \frac{3}{8} dx = \frac{3x}{8}$.

Step 9. $\int \frac{\cos 2x}{2} dx$: since $\int \cos(2x) dx = \frac{\sin 2x}{2}$, this equals $\frac{1}{2} \cdot \frac{\sin 2x}{2} = \frac{\sin 2x}{4}$.

Step 10. $\int \frac{\cos 4x}{8} dx$: since $\int \cos(4x) dx = \frac{\sin 4x}{4}$, this equals $\frac{1}{8} \cdot \frac{\sin 4x}{4} = \frac{\sin 4x}{32}$.

Step 11. Add the three results and the constant:

$$\frac{3x}{8} + \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C.$$

Step 12. Verify at one value. The derivative of the answer is $\frac{3}{8} + \frac{\cos 2x}{2} + \frac{\cos 4x}{8}$. At $x = 0$

this is $\frac{3}{8} + \frac{1}{2} + \frac{1}{8} = 1$, and $\cos^4 0 = 1$. At $x = \frac{\pi}{2}$ it is $\frac{3}{8} - \frac{1}{2} + \frac{1}{8} = 0$, and $\cos^4 \frac{\pi}{2} = 0$.

Answer. $\frac{3x}{8} + \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$

EXAMPLE 4

Level: Intermediate. Secant and tangent with odd tangent power

Evaluate $\int \sec^4 x \tan^3 x dx$.

Step 1. Identify the powers: secant has power $n = 4$ (even) and tangent has power $m = 3$ (odd). We will use the derivatives $\frac{d}{dx}[\sec x] = \sec x \tan x$ and the identity $\tan^2 x + 1 = \sec^2 x$.

Step 2. Two rules apply here. Because the tangent power is odd, we can peel off one factor $\sec x \tan x$ (which is $d(\sec x)$) and use $u = \sec x$. (The even secant power would also allow $u = \tan x$; we follow the odd-tangent rule.)

Step 3. Split off the factor: $\sec^4 x \tan^3 x = \sec^3 x \tan^2 x \cdot (\sec x \tan x)$. Check: $\sec^3 \cdot \sec = \sec^4$ and $\tan^2 \cdot \tan = \tan^3$.

Step 4. Convert the remaining even tangent power with $\tan^2 x = \sec^2 x - 1$:

$$\sec^4 x \tan^3 x = \sec^3 x (\sec^2 x - 1) (\sec x \tan x).$$

Step 5. Substitute $u = \sec x$, so $du = \sec x \tan x dx$. Every factor is now in terms of u :

$$\int u^3(u^2 - 1) du.$$

Step 6. Expand: $u^3(u^2 - 1) = u^5 - u^3$.

Step 7. Apply the power rule: $\int (u^5 - u^3) du = \frac{u^6}{6} - \frac{u^4}{4} + C$.

Step 8. Back-substitute $u = \sec x$:

$$\frac{\sec^6 x}{6} - \frac{\sec^4 x}{4} + C.$$

Step 9. Verify by differentiating with the chain rule: $\frac{d}{dx} \left[\frac{\sec^6 x}{6} \right] = \sec^5 x \cdot \sec x \tan x = \sec^6 x \tan x$ and $\frac{d}{dx} \left[\frac{\sec^4 x}{4} \right] = \sec^3 x \cdot \sec x \tan x = \sec^4 x \tan x$.

Step 10. Subtract: $\sec^6 x \tan x - \sec^4 x \tan x = \sec^4 x \tan x (\sec^2 x - 1) = \sec^4 x \tan x \cdot \tan^2 x = \sec^4 x \tan^3 x$, the original integrand.

Answer. $\frac{\sec^6 x}{6} - \frac{\sec^4 x}{4} + C$

EXAMPLE 5

Level: Advanced. Both powers even: sine squared times cosine squared

Evaluate $\int \sin^2 x \cos^2 x \, dx$.

Step 1. Identify the powers: $\sin x$ has power 2 and $\cos x$ has power 2, both even, so no factor can be peeled off for a substitution. Applying half-angle identities to both factors separately would give a product of two sums, which is messy, so first combine the factors.

Step 2. Recall the double-angle formula $\sin 2x = 2 \sin x \cos x$. Dividing by 2: $\sin x \cos x = \frac{\sin 2x}{2}$.

Step 3. Square both sides:

$$\sin^2 x \cos^2 x = (\sin x \cos x)^2 = \frac{\sin^2 2x}{4}.$$

Step 4. The new factor $\sin^2 2x$ is an even power of a single function, so use the half-angle identity $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$ with $\theta = 2x$:

$$\sin^2 2x = \frac{1 - \cos 4x}{2}.$$

Step 5. Substitute:

$$\sin^2 x \cos^2 x = \frac{1}{4} \cdot \frac{1 - \cos 4x}{2} = \frac{1 - \cos 4x}{8}.$$

Step 6. Check the identity at $x = \frac{\pi}{4}$: the left side is $\left(\frac{\sqrt{2}}{2}\right)^2 \left(\frac{\sqrt{2}}{2}\right)^2 = \frac{1}{4}$ and the right side is $\frac{1 - \cos \pi}{8} = \frac{2}{8} = \frac{1}{4}$. They agree.

Step 7. Integrate:

$$\int \frac{1 - \cos 4x}{8} \, dx = \frac{1}{8} \int 1 \, dx - \frac{1}{8} \int \cos 4x \, dx.$$

Step 8. The first term is $\frac{x}{8}$. For the second, $\int \cos 4x \, dx = \frac{\sin 4x}{4}$ (substitution $w = 4x$, $dx = \frac{dw}{4}$), so the term is $\frac{1}{8} \cdot \frac{\sin 4x}{4} = \frac{\sin 4x}{32}$.

Step 9. Combine:

$$\frac{x}{8} - \frac{\sin 4x}{32} + C.$$

Step 10. Verify by differentiating: $\frac{d}{dx} \left[\frac{x}{8} - \frac{\sin 4x}{32} \right] = \frac{1}{8} - \frac{4 \cos 4x}{32} = \frac{1 - \cos 4x}{8} =$

$\sin^2 x \cos^2 x$, as required.

Answer. $\frac{x}{8} - \frac{\sin 4x}{32} + C$

EXAMPLE 6

Level: Intermediate. Even power of secant: the substitution $u = \tan x$

Evaluate $\int \sec^4 x \tan^2 x \, dx$.

Step 1. Identify the powers: secant has power 4 (even) and tangent has power 2 (even). The tangent power is not odd, so the odd-tangent rule ($u = \sec x$) does not apply. The secant power is even, so we use the even-secant rule with $u = \tan x$.

Step 2. The reason this works: $\frac{d}{dx}[\tan x] = \sec^2 x$, so a single factor $\sec^2 x \, dx$ becomes du .

Step 3. Peel off one factor $\sec^2 x$: $\sec^4 x = \sec^2 x \cdot \sec^2 x$.

Step 4. Convert the other $\sec^2 x$ into tangents with the identity $\sec^2 x = 1 + \tan^2 x$:

$$\sec^4 x \tan^2 x = (1 + \tan^2 x) \tan^2 x \cdot \sec^2 x.$$

Step 5. Substitute $u = \tan x$, so $du = \sec^2 x \, dx$. The integral becomes $\int (1 + u^2)u^2 \, du$.

Step 6. Expand: $(1 + u^2)u^2 = u^2 + u^4$.

Step 7. Apply the power rule: $\int (u^2 + u^4) \, du = \frac{u^3}{3} + \frac{u^5}{5} + C$.

Step 8. Back-substitute $u = \tan x$:

$$\frac{\tan^3 x}{3} + \frac{\tan^5 x}{5} + C.$$

Step 9. Verify by differentiating: $\frac{d}{dx} \left[\frac{\tan^3 x}{3} \right] = \tan^2 x \sec^2 x$ and $\frac{d}{dx} \left[\frac{\tan^5 x}{5} \right] = \tan^4 x \sec^2 x$.

Step 10. Add: $\tan^2 x \sec^2 x + \tan^4 x \sec^2 x = \tan^2 x \sec^2 x (1 + \tan^2 x) = \tan^2 x \sec^2 x \cdot \sec^2 x = \sec^4 x \tan^2 x$, the original integrand.

Step 11. Remark on where $\int \sec x \, dx$ enters: when the secant power is odd and the tangent power is even (for example $\int \sec x \, dx$ itself), neither rule applies. Multiplying by $\frac{\sec x + \tan x}{\sec x + \tan x}$ gives $\int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx$, and with $u = \sec x + \tan x$, $du = (\sec x \tan x + \sec^2 x) \, dx$, the result is $\ln |\sec x + \tan x| + C$. It is not needed in this example because the secant power here is even.

Answer. $\frac{\tan^3 x}{3} + \frac{\tan^5 x}{5} + C$

EXAMPLE 7*Level: Foundational. Long division, then integrate*

Evaluate

$$\int \frac{x^2 + 3x + 5}{x + 1} dx.$$

Step 1. Compare degrees. The numerator $x^2 + 3x + 5$ has degree 2 and the denominator $x + 1$ has degree 1. Partial fractions only work when the numerator has strictly smaller degree than the denominator, so we must first divide the numerator by the denominator (long division).

Step 2. Start the division. Divide the leading term of the numerator by the leading term of the denominator: $\frac{x^2}{x} = x$. This is the first term of the quotient.

Step 3. Multiply that term back by the whole divisor: $x(x + 1) = x^2 + x$. Subtract it from the numerator: $(x^2 + 3x + 5) - (x^2 + x) = 2x + 5$.

Step 4. Repeat with the new leading term: $\frac{2x}{x} = 2$. Multiply back: $2(x + 1) = 2x + 2$. Subtract: $(2x + 5) - (2x + 2) = 3$.

Step 5. The remainder 3 has degree 0, which is less than the degree 1 of the divisor, so the division stops. The quotient is $x + 2$ and the remainder is 3, that is $x^2 + 3x + 5 = (x + 1)(x + 2) + 3$.

Step 6. Check the division by expanding: $(x + 1)(x + 2) = x^2 + 2x + x + 2 = x^2 + 3x + 2$, and adding the remainder gives $x^2 + 3x + 2 + 3 = x^2 + 3x + 5$, which is the original numerator.

Step 7. Divide both sides of $x^2 + 3x + 5 = (x + 1)(x + 2) + 3$ by $x + 1$ to rewrite the integrand:

$$\frac{x^2 + 3x + 5}{x + 1} = x + 2 + \frac{3}{x + 1}.$$

Step 8. Split the integral into three pieces:

$$\int \left(x + 2 + \frac{3}{x + 1} \right) dx = \int x dx + \int 2 dx + 3 \int \frac{dx}{x + 1}.$$

Step 9. Integrate the first two pieces with the power rule $\int x^n dx = \frac{x^{n+1}}{n+1}$ and the constant rule: $\int x dx = \frac{x^2}{2}$ and $\int 2 dx = 2x$.

Step 10. For the third piece let $u = x + 1$, so $du = dx$. Then $\int \frac{dx}{x + 1} = \int \frac{du}{u} = \ln |u| = \ln |x + 1|$, using $\int \frac{du}{u} = \ln |u| + C$.

Step 11. Assemble the result:

$$\int \frac{x^2 + 3x + 5}{x + 1} dx = \frac{x^2}{2} + 2x + 3 \ln |x + 1| + C.$$

Step 12. Verify by differentiating: $\frac{d}{dx} \left[\frac{x^2}{2} + 2x + 3 \ln |x + 1| \right] = x + 2 + \frac{3}{x + 1} = \frac{(x + 2)(x + 1) + 3}{x + 1} = \frac{x^2 + 3x + 5}{x + 1}$, which is the original integrand.

Answer. $\frac{x^2}{2} + 2x + 3 \ln |x + 1| + C$

EXAMPLE 8*Level: Intermediate. Repeated linear factor*

Evaluate

$$\int \frac{2x + 6}{(x + 1)^2} dx.$$

Step 1. The numerator has degree 1 and the denominator $(x + 1)^2$ has degree 2, so no long division is needed and we can decompose directly.

Step 2. The denominator contains the repeated linear factor $(x + 1)^2$. A repeated factor needs one term for every power from 1 up to 2, so the shape is

$$\frac{2x + 6}{(x + 1)^2} = \frac{A}{x + 1} + \frac{B}{(x + 1)^2}.$$

Step 3. Multiply both sides by the common denominator $(x + 1)^2$ to clear the fractions:

$$2x + 6 = A(x + 1) + B.$$

Step 4. Find B by choosing the value of x that kills the A term, namely $x = -1$: $2(-1) + 6 = A(0) + B$, so $B = 4$.

Step 5. Find A by comparing the coefficients of x on both sides. The right side expands to $Ax + A + B$, whose coefficient of x is A ; the left side has coefficient 2. So $A = 2$.

Step 6. Check with the constant terms: $A + B = 2 + 4 = 6$, which matches the constant 6 on the left. The decomposition is

$$\frac{2x + 6}{(x + 1)^2} = \frac{2}{x + 1} + \frac{4}{(x + 1)^2}.$$

Step 7. Integrate term by term:

$$\int \frac{2x + 6}{(x + 1)^2} dx = 2 \int \frac{dx}{x + 1} + 4 \int \frac{dx}{(x + 1)^2}.$$

Step 8. Substitute $u = x + 1$, $du = dx$ in both integrals. The first is $2 \int \frac{du}{u} = 2 \ln |u| = 2 \ln |x + 1|$.

Step 9. The second is $4 \int u^{-2} du$. By the power rule $\int u^{-2} du = \frac{u^{-1}}{-1} = -\frac{1}{u}$, so it equals $-\frac{4}{u} = -\frac{4}{x + 1}$.

Step 10. Combine:

$$\int \frac{2x + 6}{(x + 1)^2} dx = 2 \ln |x + 1| - \frac{4}{x + 1} + C.$$

Step 11. Verify by differentiating: $\frac{d}{dx} \left[2 \ln |x + 1| - \frac{4}{x + 1} \right] = \frac{2}{x + 1} + \frac{4}{(x + 1)^2} = \frac{2(x + 1) + 4}{(x + 1)^2} = \frac{2x + 6}{(x + 1)^2}$, the original integrand.

Answer. $2 \ln |x + 1| - \frac{4}{x + 1} + C$

EXAMPLE 9*Level: Intermediate. Irreducible quadratic factor*

Evaluate

$$\int \frac{3x+1}{(x^2+1)(x-2)} dx.$$

Step 1. The numerator has degree 1 and the denominator has degree 3, so we decompose directly. The factor $x-2$ is a distinct linear factor. The factor x^2+1 cannot be factored over the real numbers (it has no real roots, since $x^2+1 \geq 1$), so it is an irreducible quadratic.

Step 2. An irreducible quadratic contributes a term with a linear numerator, and a linear factor contributes a constant numerator:

$$\frac{3x+1}{(x^2+1)(x-2)} = \frac{Ax+B}{x^2+1} + \frac{C}{x-2}.$$

Step 3. Multiply both sides by $(x^2+1)(x-2)$:

$$3x+1 = (Ax+B)(x-2) + C(x^2+1).$$

Step 4. Find C by substituting $x=2$, which makes $x-2=0$: $3(2)+1 = (2A+B)(0) + C(4+1)$, so $7 = 5C$ and $C = \frac{7}{5}$.

Step 5. Expand the right side fully: $(Ax+B)(x-2) = Ax^2 - 2Ax + Bx - 2B$, so

$$3x+1 = (A+C)x^2 + (B-2A)x + (C-2B).$$

Step 6. Match coefficients of x^2 : the left side has 0, so $A+C=0$ and $A=-C=-\frac{7}{5}$.

Step 7. Match the constant terms: $C-2B=1$, so $2B=C-1=\frac{7}{5}-1=\frac{2}{5}$ and $B=\frac{1}{5}$.

Step 8. Check with the coefficient of x : $B-2A=\frac{1}{5}+\frac{14}{5}=\frac{15}{5}=3$, which matches the 3 on the left. So

$$\frac{3x+1}{(x^2+1)(x-2)} = \frac{-\frac{7}{5}x+\frac{1}{5}}{x^2+1} + \frac{\frac{7}{5}}{x-2}.$$

Step 9. Split the first fraction into two and integrate each part:

$$\int \frac{3x+1}{(x^2+1)(x-2)} dx = -\frac{7}{5} \int \frac{x}{x^2+1} dx + \frac{1}{5} \int \frac{dx}{x^2+1} + \frac{7}{5} \int \frac{dx}{x-2}.$$

Step 10. For the first integral let $u = x^2+1$, so $du = 2x dx$ and $x dx = \frac{du}{2}$. Then

$$\int \frac{x}{x^2+1} dx = \frac{1}{2} \ln|u| = \frac{1}{2} \ln(x^2+1) \text{ (no absolute value needed since } x^2+1 > 0 \text{)}.$$

Step 11. The second integral is the standard form $\int \frac{du}{u^2+a^2} = \frac{1}{a} \arctan\left(\frac{u}{a}\right)$ with $a=1$, giving $\arctan x$. The third is $\ln|x-2|$ by $u=x-2$.

Step 12. Assemble:

$$-\frac{7}{5} \cdot \frac{1}{2} \ln(x^2+1) + \frac{1}{5} \arctan x + \frac{7}{5} \ln|x-2| + C = -\frac{7}{10} \ln(x^2+1) + \frac{1}{5} \arctan x + \frac{7}{5} \ln|x-2| + C.$$

Step 13. Verify at $x=0$: the integrand is $\frac{1}{(1)(-2)} = -\frac{1}{2}$. The derivative of the answer at 0

is $-\frac{7}{10} \cdot \frac{0}{1} + \frac{1}{5} \cdot \frac{1}{1} + \frac{7}{5} \cdot \frac{1}{-2} = \frac{1}{5} - \frac{7}{10} = -\frac{1}{2}$, as required.

Answer. $-\frac{7}{10} \ln(x^2+1) + \frac{1}{5} \arctan x + \frac{7}{5} \ln|x-2| + C$

EXAMPLE 10*Level: Advanced. Definite integral with an exact logarithm*

Evaluate

$$\int_0^1 \frac{4x+5}{(x+1)(x+2)} dx$$

and give the exact value as a single logarithm.

Step 1. The numerator has degree 1 and the denominator has degree 2, and the denominator is already factored into distinct linear factors, so the shape is

$$\frac{4x+5}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}.$$

Step 2. Clear denominators by multiplying by $(x+1)(x+2)$:

$$4x+5 = A(x+2) + B(x+1).$$

Step 3. Substitute $x = -1$ (which kills the B term): $4(-1) + 5 = A(1) + B(0)$, so $A = 1$.**Step 4.** Substitute $x = -2$ (which kills the A term): $4(-2) + 5 = A(0) + B(-1)$, so $-3 = -B$ and $B = 3$.**Step 5.** Check by recombining: $1 \cdot (x+2) + 3(x+1) = x+2+3x+3 = 4x+5$, which matches. So

$$\frac{4x+5}{(x+1)(x+2)} = \frac{1}{x+1} + \frac{3}{x+2}.$$

Step 6. Find an antiderivative. Using $\int \frac{dx}{x+a} = \ln|x+a|$ (substitute $u = x+a$),

$$\int \left(\frac{1}{x+1} + \frac{3}{x+2} \right) dx = \ln|x+1| + 3\ln|x+2|.$$

Step 7. The integrand is continuous on $[0, 1]$ (the denominator vanishes only at $x = -1$ and $x = -2$), so the Fundamental Theorem of Calculus applies. On $[0, 1]$ both $x+1$ and $x+2$ are positive, so the absolute values can be dropped.**Step 8.** Evaluate at the upper limit $x = 1$: $\ln 2 + 3\ln 3$.**Step 9.** Evaluate at the lower limit $x = 0$: $\ln 1 + 3\ln 2 = 0 + 3\ln 2 = 3\ln 2$.**Step 10.** Subtract (upper minus lower): $\ln 2 + 3\ln 3 - 3\ln 2 = 3\ln 3 - 2\ln 2$.**Step 11.** Combine into one logarithm using $n \ln a = \ln a^n$ and $\ln a - \ln b = \ln \frac{a}{b}$: $3\ln 3 - 2\ln 2 = \ln 27 - \ln 4 = \ln \frac{27}{4}$.**Step 12.** Sanity check numerically: $3\ln 3 - 2\ln 2 \approx 3(1.0986) - 2(0.6931) = 3.2958 - 1.3863 = 1.9095$, and $\ln \frac{27}{4} = \ln 6.75 \approx 1.9095$. The integrand is between $\frac{5}{2}$ (at $x = 0$) and $\frac{9}{6} = 1.5$ (at $x = 1$) on an interval of length 1, so a value near 1.9 is reasonable.**Answer.** $\ln \frac{27}{4} = 3\ln 3 - 2\ln 2 \approx 1.9095$ **EXAMPLE 11***Level: Advanced. Weierstrass substitution with arctangent*

Evaluate

$$\int \frac{1}{2 + \cos x} dx.$$

Step 1. Choose the technique. The integrand is a rational function of $\cos x$ with no power

of sine or cosine to peel off and no secant or tangent structure, so the odd-power, even-power and sec/tan rules do not apply. The Weierstrass substitution $t = \tan\left(\frac{x}{2}\right)$ converts any rational function of $\sin x$ and $\cos x$ into a rational function of t .

Step 2. Derive $\cos x$ in terms of t . By the double-angle identity $\cos x = \cos^2\left(\frac{x}{2}\right) - \sin^2\left(\frac{x}{2}\right)$, and by the Pythagorean identity $1 = \cos^2\left(\frac{x}{2}\right) + \sin^2\left(\frac{x}{2}\right)$. Write $\cos x = \frac{\cos^2(x/2) - \sin^2(x/2)}{\cos^2(x/2) + \sin^2(x/2)}$ and divide numerator and denominator by $\cos^2\left(\frac{x}{2}\right)$ to get

$$\cos x = \frac{1 - \tan^2(x/2)}{1 + \tan^2(x/2)} = \frac{1 - t^2}{1 + t^2}.$$

Step 3. Find dx . From $t = \tan\left(\frac{x}{2}\right)$ we get $x = 2 \arctan t$, and since $\frac{d}{dt} \arctan t = \frac{1}{1 + t^2}$,

$$dx = \frac{2 dt}{1 + t^2}.$$

(This is valid for $-\pi < x < \pi$, where $\tan\left(\frac{x}{2}\right)$ is defined.)

Step 4. Substitute into the integral:

$$\int \frac{1}{2 + \frac{1 - t^2}{1 + t^2}} \cdot \frac{2 dt}{1 + t^2}.$$

Step 5. Simplify the denominator by writing 2 over the common denominator $1 + t^2$: $2 + \frac{1 - t^2}{1 + t^2} = \frac{2(1 + t^2) + (1 - t^2)}{1 + t^2} = \frac{2 + 2t^2 + 1 - t^2}{1 + t^2} = \frac{t^2 + 3}{1 + t^2}$.

Step 6. Dividing by a fraction means multiplying by its reciprocal, so the integrand becomes $\frac{1 + t^2}{t^2 + 3} \cdot \frac{2}{1 + t^2}$. The factor $1 + t^2$ cancels:

$$\int \frac{2 dt}{t^2 + 3}.$$

Step 7. Recognise the standard form $\int \frac{du}{u^2 + a^2} = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$ with $u = t$ and $a = \sqrt{3}$ (since $3 = (\sqrt{3})^2$).

Step 8. Apply it:

$$\int \frac{2 dt}{t^2 + 3} = \frac{2}{\sqrt{3}} \arctan\left(\frac{t}{\sqrt{3}}\right) + C.$$

Step 9. Return to the original variable by replacing $t = \tan\left(\frac{x}{2}\right)$:

$$\int \frac{1}{2 + \cos x} dx = \frac{2}{\sqrt{3}} \arctan\left(\frac{\tan(x/2)}{\sqrt{3}}\right) + C.$$

Step 10. Verify by differentiating with respect to x . Use $\frac{dt}{dx} = \frac{1}{2} \sec^2\left(\frac{x}{2}\right) = \frac{1 + t^2}{2}$ and the chain rule:

$$\frac{d}{dx} \left[\frac{2}{\sqrt{3}} \arctan\left(\frac{t}{\sqrt{3}}\right) \right] = \frac{2}{\sqrt{3}} \cdot \frac{1}{1 + t^2/3} \cdot \frac{1}{\sqrt{3}} \cdot \frac{1 + t^2}{2} = \frac{1 + t^2}{3 + t^2}.$$

Step 11. Compare with the integrand: $\frac{1}{2 + \cos x} = \frac{1 + t^2}{3 + t^2}$ from step 5 (the reciprocal of

$\frac{t^2 + 3}{1 + t^2}$). The two agree, so the antiderivative is correct. The substitution turned an integral with no obvious pattern into a routine arctangent integral.

Answer. $\frac{2}{\sqrt{3}} \arctan \left(\frac{\tan(x/2)}{\sqrt{3}} \right) + C$

EXAMPLE 12

Level: Challenge. Weierstrass substitution leading to a logarithm

Evaluate

$$\int \frac{1}{4 + 5 \sin x} dx.$$

Step 1. The integrand is a rational function of $\sin x$ alone with no power to peel off, so use the Weierstrass substitution $t = \tan \left(\frac{x}{2} \right)$, for which $\sin x = \frac{2t}{1 + t^2}$ and $dx = \frac{2 dt}{1 + t^2}$. (The identity for sine comes from $\sin x = 2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)$ divided by $1 = \cos^2 \left(\frac{x}{2} \right) + \sin^2 \left(\frac{x}{2} \right)$, then dividing top and bottom by $\cos^2 \left(\frac{x}{2} \right)$.)

Step 2. Rewrite the denominator over the common denominator $1 + t^2$:

$$4 + 5 \sin x = 4 + \frac{10t}{1 + t^2} = \frac{4(1 + t^2) + 10t}{1 + t^2} = \frac{4t^2 + 10t + 4}{1 + t^2}.$$

Step 3. Substitute into the integral and take the reciprocal of the denominator:

$$\int \frac{1 + t^2}{4t^2 + 10t + 4} \cdot \frac{2 dt}{1 + t^2} = \int \frac{2 dt}{4t^2 + 10t + 4}.$$

Step 4. Cancel the factor $1 + t^2$ and divide numerator and denominator by 2:

$$\int \frac{dt}{2t^2 + 5t + 2}.$$

Step 5. Decide arctangent or logarithm by the discriminant of $2t^2 + 5t + 2$: $b^2 - 4ac = 25 - 16 = 9 > 0$. It is positive, so the quadratic factors over the real numbers and we expect a logarithm rather than an arctangent.

Step 6. Factor: $2t^2 + 5t + 2 = (2t + 1)(t + 2)$. Check by expanding: $(2t + 1)(t + 2) = 2t^2 + 4t + t + 2 = 2t^2 + 5t + 2$.

Step 7. Set up partial fractions:

$$\frac{1}{(2t + 1)(t + 2)} = \frac{A}{2t + 1} + \frac{B}{t + 2}, \quad \text{so} \quad 1 = A(t + 2) + B(2t + 1).$$

Step 8. Substitute $t = -2$: $1 = A(0) + B(-4 + 1) = -3B$, so $B = -\frac{1}{3}$.

Step 9. Substitute $t = -\frac{1}{2}$: $1 = A \left(\frac{3}{2} \right) + B(0)$, so $A = \frac{2}{3}$. Check by recombining: $\frac{2}{3}(t + 2) - \frac{1}{3}(2t + 1) = \frac{2t + 4 - 2t - 1}{3} = 1$.

Step 10. Integrate, using $\int \frac{dt}{at + b} = \frac{1}{a} \ln |at + b|$:

$$\int \left(\frac{2/3}{2t + 1} - \frac{1/3}{t + 2} \right) dt = \frac{2}{3} \cdot \frac{1}{2} \ln |2t + 1| - \frac{1}{3} \ln |t + 2| = \frac{1}{3} \ln |2t + 1| - \frac{1}{3} \ln |t + 2|.$$

Step 11. Combine with $\ln a - \ln b = \ln \frac{a}{b}$ and substitute back $t = \tan\left(\frac{x}{2}\right)$:

$$\int \frac{dx}{4 + 5 \sin x} = \frac{1}{3} \ln \left| \frac{2 \tan(x/2) + 1}{\tan(x/2) + 2} \right| + C.$$

Step 12. Verify by differentiating. With $\frac{dt}{dx} = \frac{1+t^2}{2}$: $\frac{d}{dx} \left[\frac{1}{3} \ln \left| \frac{2t+1}{t+2} \right| \right] = \frac{1}{3} \left(\frac{2}{2t+1} - \frac{1}{t+2} \right) \frac{1+t^2}{2} = \frac{1}{(2t+1)(t+2)} \cdot \frac{1+t^2}{2} = \frac{1+t^2}{4t^2+10t+4}$, and this equals $\frac{1}{4+5 \sin x}$ by step 2.

Answer. $\frac{1}{3} \ln \left| \frac{2 \tan(x/2) + 1}{\tan(x/2) + 2} \right| + C$

Practice Problems

Trigonometric Integrals

1. Evaluate (a) $\int \sin^{10} x \cos x \, dx$ and (b) $\int \sin^{10} x \cos^7 x \, dx$. Identify which technique each needs.

Your working:

2. Evaluate $\int \sec^2 x \tan^3 x \, dx$.

Your working:

3. Evaluate $\int \tan^3 x \, dx$.

Your working:

4. Evaluate $\int_0^{\pi/2} \cos^3 x \, dx$.

Your working:

5. Evaluate $\int_0^\pi \cos^2 x \, dx$.

Your working:

6. Evaluate $\int \tan^4 x \, dx$.

Your working:

7. Evaluate $\int \sin^4 x \, dx$. (Both powers are even, so apply the half-angle identity twice.)

Your working:

8. State which substitution ($u = \tan x$ or $u = \sec x$) is appropriate for $\int \sec^5 x \tan^3 x \, dx$ and explain why, using the rules for $\int \sec^n x \tan^m x \, dx$. Then evaluate the integral.

Your working:

9. Evaluate $\int \sin^2 x \cos^4 x \, dx$.

Your working:

10. A student claims that $\int \sin^3 x \, dx = -\cos x - \frac{\cos^3 x}{3} + C$, using the step $\sin^3 x = \sin x (1 + \cos^2 x)$. Find the error and give the correct result.

Your working:

Partial Fractions

11. Factor the denominator completely and evaluate

$$\int \frac{1}{x^2 + 3x} \, dx.$$

Your working:

12. Factor the denominator completely and evaluate

$$\int \frac{1}{x^3 - x} dx.$$

Your working:

13. Evaluate

$$\int \frac{5x + 1}{2x^2 + 5x - 3} dx.$$

Your working:

14. A student evaluates $\int \frac{3x+1}{(x-1)(x+2)} dx$ as follows. They write $\frac{3x+1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$, then use the cover-up shortcut: at $x = 1$ they compute $A = 3(1) + 1 = 4$, and at $x = -2$ they compute $B = 3(-2) + 1 = -5$. They conclude that the integral equals $4 \ln |x-1| - 5 \ln |x+2| + C$. Find the error and give the correct result.

Your working:

15. Evaluate the definite integral

$$\int_2^3 \frac{1}{x^2 - 1} dx$$

and give the exact value as a single logarithm.

Your working:

16. Evaluate

$$\int \frac{3x^2 + 2}{x^2 + 1} dx.$$

Your working:

17. Evaluate

$$\int \frac{x + 1}{x^2(x - 2)} dx.$$

Your working:

18. Evaluate

$$\int \frac{2x}{(x^2 + 1)(x - 1)} dx.$$

Your working:

19. Evaluate

$$\int \frac{x^4 + 3x^2 - x + 1}{(x - 1)(x^2 + 1)} dx.$$

Your working:

The Weierstrass Substitution

20. Use $t = \tan\left(\frac{x}{2}\right)$ to evaluate

$$\int \frac{1}{1 + \sin x} dx.$$

Your working:

21. Use $t = \tan\left(\frac{x}{2}\right)$ to evaluate

$$\int_0^{\pi/2} \frac{1}{1 + \sin x + \cos x} dx.$$

Your working:

22. Use $t = \tan\left(\frac{x}{2}\right)$ to evaluate

$$\int \frac{1}{3 + 5 \cos x} dx,$$

and simplify the resulting expression fully.

Your working:

23. Use $t = \tan\left(\frac{x}{2}\right)$ to evaluate

$$\int \frac{1}{2 + \sin x} dx.$$

Your working:

24. Show that $\int \sec x \, dx$ can be evaluated with the substitution $t = \tan\left(\frac{x}{2}\right)$, and confirm that the result agrees with the memorised formula $\ln |\sec x + \tan x| + C$.

Your working:

Mixed Technique Identification

25. For each integral, state which single technique from Weeks 3 and 4 (substitution, integration by parts, trig-power identities, partial fractions, or the Weierstrass substitution) is the most efficient, explain why, and then evaluate it: (a) $\int \frac{x}{x^2 - 9} \, dx$; (b) $\int x^2 \cos(x^3) \, dx$; (c) $\int \frac{1}{\sin x - \cos x} \, dx$; (d) $\int \sec^6 x \, dx$.

Your working:

26. For each integral, name the most efficient single technique (substitution, integration by parts, trig-power identities, partial fractions, or the Weierstrass substitution) and then evaluate it: (a) $\int x \cos x \, dx$; (b) $\int \frac{1}{x^2 - 4x + 3} \, dx$; (c) $\int \cos^3 x \, dx$.

Your working:
